

`TaperedPauliSumOp` is incompatible with `VQE`

<https://github.com/Qiskit/qiskit/issues/6100>

ROW 91

TaperedPauliSumOp is incompatible with VQE #6100

[New issue](#)[Closed](#)

#6102



mrossinek opened on Mar 27, 2021

Information

- Qiskit Terra version: master @ @c8bb3d
- Python version: 3.8.8
- Operating system: Linux (Fedora 33): uname -r: 5.10.23-200.fc33.x86_64

What is the current behavior?

Using a tapered `PauliSumOp` as the input to a `VQE` results in the following runtime error:

```
Traceback (most recent call last):
  File "terra-bug-minimal.py", line 31, in <module>
    result = vqe.compute_minimum_eigenvalue(tapered_qubit_op)
  File "/home/oss/Files/Qiskit/src/qiskit-terra/qiskit/algorithms/minimum_eigen_solvers/vqe.py", line 412, in
    self._expect_op = self.construct_expectation(self._var_form_params, operator)
  File "/home/oss/Files/Qiskit/src/qiskit-terra/qiskit/algorithms/minimum_eigen_solvers/vqe.py", line 300, in
    observable_meas = self.expectation.convert(StateFn(operator, is_measurement=True))
  File "/home/oss/Files/Qiskit/src/qiskit-terra/qiskit/opflow/expectations/matrix_expectation.py", line 38, in
    return operator.to_matrix_op()
  File "/home/oss/Files/Qiskit/src/qiskit-terra/qiskit/opflow/state_fns/operator_state_fn.py", line 119, in to
    return OperatorStateFn(self.primitive.to_matrix_op(massive=massive) * self.coeff,
  File "/home/oss/Files/Qiskit/src/qiskit-terra/qiskit/opflow/primitive_ops/primitive_op.py", line 244, in to
    prim_mat = self.__class__(self.primitive).to_matrix(massive=massive)
TypeError: __init__() missing 1 required positional argument: 'z2_symmetries'
```

Steps to reproduce the problem

Below is a minimal code snippet to reproduce the issue:

```
from qiskit import BasicAer
from qiskit.algorithms import VQE
from qiskit.circuit.library import TwoLocal
from qiskit.opflow import PauliSumOp, TwoQubitReduction

qubit_op = PauliSumOp.from_list([
    ("IIII", -0.8105479805373266),
    ("IIIZ", 0.17218393261915552),
    ("IIZZ", -0.22575349222402472),
    ("IIZI", 0.17218393261915556),
    ("ZZII", -0.22575349222402466),
    ("IIZI", 0.1209126326177663),
    ("IIZZ", 0.16892753870087912),
    ("IXZX", -0.045232799946057854),
    ("ZXIX", 0.045232799946057854),
    ("IXIX", 0.045232799946057854),
    ("ZXZX", -0.045232799946057854),
    ("ZZIZ", 0.16614543256382414),
    ("IIZI", 0.16614543256382414),
    ("ZZZZ", 0.17464343068300453),
    ("IIZI", 0.1209126326177663),
])

tapered_qubit_op = TwoQubitReduction(num_particles=2).convert(qubit_op)

backend = BasicAer.get_backend('statevector_simulator')
ryrz = TwoLocal(rotation_blocks=['ry', 'rz'], entanglement_blocks=['cz'])

vqe = VQE(ryrz, quantum_instance=backend)

result = vqe.compute_minimum_eigenvalue(tapered_qubit_op)
```

What is the expected behavior?

I would expect that the `VQE` is able to compute the minimum eigenvalue just like for any other `opflow` operator type.

Assignees

No one assigned

Labels

`bug` `mod: opflow`

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

[Fix TaperedPauliSumOp.to_matrix_op\(\)](#)
Qiskit/qiskit

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Participants



What is the expected behavior?

I would expect that the `VQE` is able to compute the minimum eigenvalue just like for any other `opflow` operator type.

Suggested solutions

A simple solution would be to make the `z2_symmetries` argument of `TaperedPauliSumOp.__init__` optional. However, this may lead to other unwanted side-effects.

Another possible solution would be to make the `TaperedPauliSumOp` look more like a `PauliSumOp` (at least in this context). But I am unsure how to achieve this given the `__init__` call which is raising the problem is obtained via `self.__class__`. And I don't think we should hack the behavior of this internal object.



mrossinek added `bug` on Mar 27, 2021



mrossinek on Mar 27, 2021

Member Author ...

[@ikkoham](#) I am tagging you because (if I recall correctly) you mainly worked on the two-qubit-reduction and tapering code.



ikkoham added a commit that references this issue on Mar 29, 2021

Fix [Qiskit#6100](#)

d6ae8b7

ikkoham mentioned this on Mar 29, 2021

Fix `TaperedPauliSumOp.to_matrix_op()` [#6102](#)



ikkoham on Mar 29, 2021

Contributor ...

Thank you for finding the bug. I have checked only `qasm_simulator`.



woodsp-ibm closed this as `completed` in [#6102](#) on Mar 29, 2021

woodsp-ibm added a commit that references this issue on Mar 29, 2021

Fix `TaperedPauliSumOp.to_matrix_op()` [\(#6102\)](#) ...

`Verified` 1986c95

mrossinek mentioned this on Mar 31, 2021

[More incompatibilities of TaperedPauliSumOp](#) [#6127](#)

woodsp-ibm added `mod: opflow` on Apr 23, 2021

ElePT added a commit that references this issue on Jun 27, 2023

Fix `TaperedPauliSumOp.to_matrix_op()` [\(Qiskit#6102\)](#) ...

3f5ac05

ElePT added a commit that references this issue on Jul 17, 2023

Fix `TaperedPauliSumOp.to_matrix_op()` [\(Qiskit/qiskit#6102\)](#) ...

a06501c

Fix TaperedPauliSumOp.to_matrix_op() #6102

<> Code

Merged woodsp-ibm merged 2 commits into `Qiskit:master` from `ikkoham:fix-6100` on Mar 29, 2021

Conversation 2 Commits 2 Checks 0 Files changed 3

+45 -4



ikkoham commented on Mar 29, 2021

Contributor

Summary

Fix #6100.

Details and comments

The arguments required for initialization may change in the operator in the future. So I fixed `PrimitiveOp.to_matrix_op` not to use the constructor.



Fix `Qiskit#6100`

d6ae8b7

ikkoham requested review from manaelmarques, woodsp-ibm and a team as code owners 4 years ago



Cryoris commented on Mar 29, 2021

Contributor

Hi @ikkoham could you add a short test to verify the fix?



1

add tests

bac95be



mrossinek approved these changes on Mar 29, 2021

View reviewed changes

mrossinek left a comment

Member

Thanks, @ikkoham, for the fast response! I also validated my use case and it now works as expected 🙌



Reviewers

	woodsp-ibm	✓
	mrossinek	✓
	manaelmarques	●

Assignees

No one assigned

Labels

Changelog: Bugfix mod: opflow

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

🔗 TaperedPauliSumOp is incompatible with VQE

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5 participants



More incompatibilities of TaperedPauliSumOp #6127

[New issue](#)[Closed](#)

#6315



mrossinek opened on Mar 31, 2021 · edited by mrossinek

Edits ...

Information

- Qiskit Terra version: master @ [28c543a](#)
- Python version: 3.8.8
- Operating system: Linux, Fedora 33: `uname -r: 5.11.10-200.fc33.x86_64`

What is the current behavior?

`TaperedPauliSumOp` cannot be multiplied with another `OperatorBase`.

Steps to reproduce the problem

Try the following snippet:

```
from qiskit.circuit import Parameter
from qiskit.opflow import PauliSumOp, TwoQubitReduction
from qiskit.opflow.evolutions import PauliTrotterEvolution

qubit_op = PauliSumOp.from_list([
    ("IIII", -0.8105479805373266),
    ("IIIZ", 0.17218393261915552),
    ("IIZZ", -0.22575349222402472),
    ("IZZI", 0.17218393261915556),
    ("ZZII", -0.22575349222402466),
    ("IIZI", 0.1209126326177663),
    ("IIZZ", 0.16892753870087912),
    ("IXZX", -0.045232799946057854),
    ("ZXIX", 0.045232799946057854),
    ("IXIX", 0.045232799946057854),
    ("ZXZX", -0.045232799946057854),
    ("ZZIZ", 0.16614543256382414),
    ("IZIZ", 0.16614543256382414),
    ("ZZZZ", 0.17464343068300453),
    ("ZIZI", 0.1209126326177663),
])

tapered_qubit_op = TwoQubitReduction(num_particles=2).convert(qubit_op)

param = Parameter('c')
param * tapered_qubit_op
```

What is the expected behavior?

The multiplication should work just fine.

Suggested solutions

Since this is very similar error to [#6100](#) I suggest to find a more general solution which makes `TaperedPauliSumOp` generally compatible. I am quite positive that otherwise more such problems will arise.

 mrossinek added [bug](#) on Mar 31, 2021

Assignees

No one assigned

Labels

[bug](#) [mod: opflow](#)

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

[Support multiplication of TaperedPauliSumOp with ParameterExpression](#)
Qiskit/qiskit

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ikkoham on Apr 1, 2021

Contributor ...

Thanks.
I am sorry. I didn't think TaperedPauliSumOp with parameter exists.
I recognized TaperedPauliSumOp.assign_parameters() also doesn't work.

My question: Is Z2_symmetry invariant under parameter multiplication?



mrossinek on Apr 1, 2021

Member Author ...

I would assume so 😊 I found this bug while trying to combine a TaperedPauliSumOp with our new EvolvedOperatorAnsatz in Nature. There, an error occurred on the following line: https://github.com/Qiskit/qiskit-nature/blob/master/qiskit_nature/circuit/library/ansatzes/evolved_operator_ansatz.py#L162

A more extensive example which should work once this issue is resolved is given in this test: https://github.com/Qiskit/qiskit-nature/blob/master/test/algorithms/ground_state_solvers/test_advanced_ucc_variants.py#L75

@pbark @woodsp-ibm Please correct me, if I am wrong about the invariance under parameter multiplication..



ikkoham on Apr 1, 2021

Contributor ...

If invariant, `parm * tapered_qubit_op` can be TaperedPauliSumOp which has same z2_symmetries. If not, `parm * tapered_qubit_op` must be PauliSumOp.



mrossinek mentioned this on Apr 23, 2021

🔗 Two Qubit Reduction sample with UCCSD fails with missing Z2_symmetries argument [qiskit-community/qiskit-nature#153](https://github.com/qiskit-community/qiskit-nature/issues/153)

woodsp-ibm added `mod: opflow` on Apr 23, 2021

ikkoham added a commit that references this issue on Apr 28, 2021

fix [Qiskit#6127](https://github.com/Qiskit/qiskit/pull/6127)

e18ea58

ikkoham mentioned this on Apr 28, 2021

🔗 Support multiplication of TaperedPauliSumOp with ParameterExpression #6315

🔒 mergify closed this as **completed** in #6315 on May 4, 2021

mergify added 3 commits that reference this issue on May 4, 2021

Support multiplication of TaperedPauliSumOp with ParameterExpression ...

Verified 17d114e

Support multiplication of TaperedPauliSumOp with ParameterExpression ...

09881fc

Support multiplication of TaperedPauliSumOp with ParameterExpression ...

Verified 2f74455